COURSE SYLLABUS

AGENTIC CODING

PHASE 1 — FOUNDATIONS

Get comfortable in the developer environment and with containerized workflows before writing any application code.

WEEKS 1-6 · MODULES 00-02 · 27 LABS · MON + WED, 6:00-8:00 PM

Phase 1 is the foundation of Agentic Coding at Temple College Workforce Development, delivered in collaboration with ChiefOS. Over six weeks you master VS Code, the Ubuntu terminal, and Docker — the tools every later module depends on. Content lives in **D2L Brightspace**; progress is gated by completing labs, not by attendance. Twice-weekly live blocks support you while you move at your own pace.

HOW PHASE 1 WORKS

1
WATCH
Every lesson opens with a 2–3 minute concept video that sets up what you're about to do.

2
DO
Every lesson has a hands-on lab — real commands on your own machine, building real things.

3
SUBMIT
Each lab produces an artifact in D2L. Completing it unlocks the next lesson.

4
GET UNSTUCK LIVE
Monday instruction + Wednesday lab clinic. Labs — not attendance — gate your progress.

WHAT YOUR COMPUTER NEEDS

Everyone: a 64-bit laptop with **8 GB RAM** (16 GB recommended), **30 GB free disk space**, reliable internet, and **administrator rights** to install software. **Windows 10/11 with WSL2 (Ubuntu)** or **macOS 12+**. No GPU required — AI models (Phase 3) run on the college's shared server.

Not supported: Chromebooks and tablets (including iPads). Phase 1 lives in a real Linux/Unix shell — you need a laptop you can install software on. Windows students may alternatively work entirely over SSH into the college Linux server.

6	3	27	2
WEEKS	MODULES	HANDS-ON LABS	LIVE BLOCKS / WEEK

SUGGESTED SIX-WEEK PACE

This schedule orients you and sets each week's live-block focus — it is a **suggested pace**, not a lock. Highlighted rows are **module checkpoints** (phase-gates) that must be passed to advance.

PHASE 1 FOUNDATIONS

Weeks 1–6 · Modules 00–02 · 27 labs

WK	MODULE	FOCUS
1	**00 · VS Code**	What an IDE is, installing VS Code, UI tour, the integrated terminal, extensions
2	**00 → 01**	Remote connections (WSL / SSH), **Module 00 checkpoint**; terminal setup + navigation
3	**01 · Terminal**	Creating files & folders, reading files, sudo & updates, installing software, permissions
4	**01 · Terminal**	Networking, processes & services, pipes & grep, users & groups, **Module 01 checkpoint**
5	**02 · Docker**	Docker install, your first container (hello-world), running a real image (nginx)
6	**02 · Docker**	Images vs containers, the VS Code Docker sidebar, **Module 02 / Phase 1 checkpoint** + buffer

LIVE-BLOCK CALENDAR

Monday and Wednesday, **6:00–8:00 PM**. Monday introduces the week's objects; Wednesday is a lab clinic. Come whenever you can — attendance does not gate progress.

WK	MONDAY — INSTRUCTION	WEDNESDAY — LAB CLINIC
1	Course kickoff; editor vs terminal vs IDE; VS Code install, UI tour, integrated terminal (00:00–02)	Clinic on labs 00:00–04 — extension installs; Windows students: pre-install WSL at home (reboot)
2	Remote connections: WSL (Windows) and SSH to the college server (00:05–06); walk Module 00 gate live	Clinic — gate submissions (00:07), key/permission errors; start terminal labs 01:00–01
3	Terminal core: making things, reading/editing files, sudo + apt, installing software, permissions (01:02–06)	Clinic — `rm -r` fear, nano/less exits, `chmod` / `chown`; budget most of clinic for permissions
4	Terminal advanced: networking, processes/services, pipes/grep, users/groups (01:07–10); demo intentional `usermod` failure	Clinic + **Module 01 phase-gate** (01:11) — timed, no-help run; redo-the-lesson triage
5	Docker: what it is, Engine install, hello-world, nginx on port 8080 (02:00–03)	Clinic — WSL `systemctl` vs `service docker start`, docker-group, port-forward paths
6	Images vs containers, cleanup, VS Code Docker sidebar (02:04–05); walk the Phase 1 gate end to end	**Phase 1 gate clinic** (02:06) + buffer — screenshot submissions, catch-up; preview Phase 2

MODULES AT A GLANCE

MODULE 00 · 8 LESSONS

VS Code & IDE

Editor vs terminal vs IDE, UI, terminal pane, extensions, Remote — WSL and SSH.

MODULE 01 · 12 LESSONS

Ubuntu Terminal

Navigate, create, inspect; sudo & apt; permissions; networking; processes; pipes & grep; users.

MODULE 02 · 7 LESSONS

Docker

Install Engine, hello-world, nginx, images vs containers, VS Code Docker sidebar, phase gate.

LAB CATALOG

Each lesson is one learning object with a paired lab. Highlighted rows are **final checkpoints** required to advance. Full lab definitions (Objective → Task → Submit → Pass criteria) live in each lesson in D2L / the course repo.

00 VS CODE & IDE FUNDAMENTALS 8 labs

#	LESSON	LAB TASK
00	What is an IDE?	Install VS Code; submit a screenshot of the welcome screen
01	VS Code UI Tour	Open a folder, use the command palette, set font size to 16
02	Integrated Terminal	Open the terminal and run `echo` commands successfully
03	Extensions Overview	Install the Python extension by Microsoft
04	Docker Extension	Install the Docker extension; sidebar populates after Module 02
05	Connecting to WSL	Connect to WSL and run `uname -a` showing Linux
06	Connecting over SSH	Connect to the host; run `hostname` and `whoami`
07	Final Checkpoint	Screenshot of a remote terminal with `whoami` and `pwd`

01 UBUNTU TERMINAL FUNDAMENTALS 12 labs · continued on next page

#	LESSON	LAB TASK
00	Setup Check	Run version, network, and sudo checks; confirm all pass
01	Where Am I?	Move between directories; use `cd -` and `cd ~`; list root dirs
02	Making Things	Create `~/docker-course/` with notes, practice, projects folders
03	Looking Inside Files	Create a notes file, append lines, edit with nano, verify 9 lines
04	Sudo and Updates	Run `sudo apt update` and `sudo apt upgrade -y` successfully
05	Installing Software	Install curl, wget, git, tree, etc.; verify each version
06	Permissions Drills	Create a 755 executable script; change ownership with `chown`

01 UBUNTU TERMINAL FUNDAMENTALS

continued · labs 07–11

#	LESSON	LAB TASK
07	Networking Drills	Ping 8.8.8.8, reach hub.docker.com, check ports with `ss`
08	Processes and Services	Run `ps` / `top` / `htop`; confirm SSH and cron services active
09	Pipes and Grep	Use pipes to count items; grep files and logs for patterns
10	User and Group Setup	Check `id` / `groups`; preview `usermod -aG docker`
11	Final Checkpoint	Run 33 commands from memory across all topics; all checks pass

02 DOCKER FUNDAMENTALS

7 labs · Phase 1 gate

#	LESSON	LAB TASK
00	Setup Check	Verify Linux, sudo, prerequisites, Module 01 complete
01	Install Docker Engine	Run Docker without sudo; `docker info` works
02	Your First Container	Run hello-world; list images and containers with `docker ps`
03	Running a Real Image	Run nginx on port 8080; load the page and view logs
04	Images vs Containers	Run three nginx containers from one image
05	VS Code Docker Sidebar	Run, stop, remove a container via the UI; view logs
06	Final Checkpoint	Run nginx, curl it, screenshot the sidebar, clean up — **Phase 1 gate**

PHASE 1 OUTCOME

By the end of Phase 1 you are **confident in the developer environment and containerized workflows** — ready for Phase 2 (FastAPI + Git/GitHub) and Phase 3 (AI-assisted development with Continue.dev on the college's shared inference server).